

```

mysql> use cg_mysql;
Database changed
mysql> /* Date/Time functions:
/*> 01. curdate() = current date
/*> 02. curtime() = current time
/*> 03. now() = current date and time
/*> 04. date() = Extract date part
/*> 05. time() = Extract time part
/*> 06. year() = extract year
/*> 07. month() = extract month
/*> 08. day()/dayofmonth() = extract day of the month
/*> 09. week() = extract week number
/*> 10. dayofweek() = extract day of the week
/*> 11. date_add() = add time to a date
/*> 12. date_sub() = subtract time from date
/*> 13. datediff() = difference between two dates
/*> 14. timestampdiff() = difference between two datetime values
/*> */
mysql> select curdate() as current__date;
+-----+
| current__date |
+-----+
| 2026-09-16    |
+-----+
1 row in set (0.04 sec)

mysql> select curtime() as current__time;
+-----+
| current__time |
+-----+
| 14:20:54      |
+-----+
1 row in set (0.00 sec)

mysql> select now() as current_date_time;
+-----+
| current_date_time |
+-----+
| 2026-09-16 14:21:27 |
+-----+
1 row in set (0.04 sec)

mysql> select date(now()) as extracted_date;
+-----+
| extracted_date |
+-----+
| 2026-09-16    |
+-----+
1 row in set (0.00 sec)

```

```
mysql> select time(now()) as extracted_time;
```

```
+-----+  
| extracted_time |  
+-----+  
| 14:23:49      |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select year(now()) as extracted_year;
```

```
+-----+  
| extracted_year |  
+-----+  
|          2026 |  
+-----+
```

```
1 row in set (0.05 sec)
```

```
mysql> select month(now()) as extracted_month;
```

```
+-----+  
| extracted_month |  
+-----+  
|              9 |  
+-----+
```

```
1 row in set (0.04 sec)
```

```
mysql> select week(now()) as extracted_week;
```

```
+-----+  
| extracted_week |  
+-----+  
|              37 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select day(now()) as extracted_day;
```

```
+-----+  
| extracted_day |  
+-----+  
|             16 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> select dayofweek(now()) as extracted_day_of_week;
```

```
+-----+  
| extracted_day_of_week |  
+-----+  
|              4 |  
+-----+
```

```
1 row in set (0.05 sec)
```

```
mysql> select date_add(curdate(), interval 100 day) as new_date;
```

```
+-----+
```

```

| new_date |
+-----+
| 2026-12-25 |
+-----+
1 row in set (0.05 sec)

```

```

mysql> select date_sub(curdate(), interval 100 day) as new_date;
+-----+
| new_date |
+-----+
| 2026-06-08 |
+-----+
1 row in set (0.00 sec)

```

```

mysql> select datediff(now(),'2004-04-13') as days_diff;
+-----+
| days_diff |
+-----+
|      8191 |
+-----+
1 row in set (0.04 sec)

```

```

mysql> select str_to_date('2024-03-02','%Y-%m-%d') as formatted_date;
+-----+
| formatted_date |
+-----+
| 2024-03-02 |
+-----+
1 row in set (0.04 sec)

```

```

mysql> select str_to_date('2024-03-02','%d-%m-%Y') as formatted_date;
+-----+
| formatted_date |
+-----+
| NULL |
+-----+
1 row in set, 1 warning (0.00 sec)

```

```

mysql> select str_to_date('20-03-2024','%d-%m-%Y') as formatted_date;
+-----+
| formatted_date |
+-----+
| 2024-03-20 |
+-----+
1 row in set (0.00 sec)

```

```

mysql> select date_format(curdate(),'%W,%M,%d,%Y') as formatted_date;
+-----+
| formatted_date |
+-----+

```

```
| Wednesday, September, 16, 2026 |  
+-----+  
1 row in set (0.02 sec)
```

```
mysql> select last_day(now()) as last_day_of_month;  
+-----+  
| last_day_of_month |  
+-----+  
| 2026-09-30        |  
+-----+  
1 row in set (0.03 sec)
```

```
mysql> select makedate(2024,60) as result_date;  
+-----+  
| result_date |  
+-----+  
| 2024-02-29  |  
+-----+  
1 row in set (0.04 sec)
```

```
mysql> select makedate(2025,60) as result_date;  
+-----+  
| result_date |  
+-----+  
| 2025-03-01  |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> select maketime(14,25,12) as result_time;  
+-----+  
| result_time |  
+-----+  
| 14:25:12    |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> select monthname(now()) as month_name;  
+-----+  
| month_name |  
+-----+  
| September  |  
+-----+  
1 row in set (0.00 sec)
```

```
mysql> select dayname(now()) as day_name;  
+-----+  
| day_name |  
+-----+  
| Wednesday |  
+-----+
```

1 row in set (0.03 sec)

```
mysql> select quarter(now()) as quarter_number;
```

quarter_number
3

1 row in set (0.00 sec)

```
mysql> select sec_to_time(3661) as time_value;
```

time_value
01:01:01

1 row in set (0.04 sec)

```
mysql> select time_to_sec(01:01:01) as time_value;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ':01:01) as time_value' at line 1

```
mysql> select time_to_sec('01:01:01') as time_value;
```

time_value
3661

1 row in set (0.05 sec)

```
mysql> CREATE TABLE employee_attendance
```

```
-> (  
-> emp_id INT PRIMARY KEY AUTO_INCREMENT,  
-> emp_name VARCHAR(100),  
-> department VARCHAR(50),  
-> joining_date DATE,  
-> birth_date DATE,  
-> login_time DATETIME,  
-> logout_time DATETIME,  
-> salary DECIMAL(10,2)  
-> );
```

Query OK, 0 rows affected (0.08 sec)

```
mysql>
```

```
mysql> INSERT INTO employee_attendance
```

```
-> (emp_name, department, joining_date, birth_date, login_time, logout_time,  
salary)
```

```
-> VALUES
```

```
-> ('Rahul', 'IT', '2021-06-15', '1998-05-10', '2024-11-20 09:05:10', '2024-11-20  
18:15:30', 55000),
```

```

-> ('Sneha','HR','2020-03-21','1996-11-18','2024-11-21 08:55:20','2024-11-21
17:45:50',48000),
-> ('Arjun','Finance','2019-08-10','1994-07-30','2024-11-22
09:15:45','2024-11-22 18:35:20',62000),
-> ('Priya','IT','2022-01-05','2000-01-25','2024-11-23 09:00:00','2024-11-23
17:55:40',51000),
-> ('Kiran','Admin','2018-12-11','1992-09-15','2024-11-24 08:40:30','2024-11-24
17:30:00',45000),
-> ('Anjali','Testing','2023-04-17','1999-12-20','2024-11-25
09:12:10','2024-11-25 18:05:00',47000),
-> ('Vijay','IT','2021-09-14','1997-04-08','2024-11-26 09:08:15','2024-11-26
18:20:40',59000),
-> ('Divya','HR','2019-10-19','1995-06-11','2024-11-27 08:50:30','2024-11-27
17:42:15',50000),
-> ('Ramesh','Finance','2017-07-01','1991-08-14','2024-11-28
09:03:00','2024-11-28 18:12:00',70000),
-> ('Pooja','Testing','2022-05-09','2001-03-19','2024-11-29
09:18:50','2024-11-29 18:25:10',53000);
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```
mysql>
```

```
mysql>
```

```
mysql> select*from employee_attendance;
```

```

+-----+-----+-----+-----+-----+-----+
+-----+-----+
| emp_id | emp_name | department | joining_date | birth_date | login_time
| logout_time | salary |
+-----+-----+-----+-----+-----+-----+
+-----+-----+
|      1 | Rahul   | IT         | 2021-06-15   | 1998-05-10 | 2024-11-20 09:05:10
| 2024-11-20 18:15:30 | 55000.00 |
|      2 | Sneha   | HR         | 2020-03-21   | 1996-11-18 | 2024-11-21 08:55:20
| 2024-11-21 17:45:50 | 48000.00 |
|      3 | Arjun   | Finance    | 2019-08-10   | 1994-07-30 | 2024-11-22 09:15:45
| 2024-11-22 18:35:20 | 62000.00 |
|      4 | Priya   | IT         | 2022-01-05   | 2000-01-25 | 2024-11-23 09:00:00
| 2024-11-23 17:55:40 | 51000.00 |
|      5 | Kiran   | Admin      | 2018-12-11   | 1992-09-15 | 2024-11-24 08:40:30
| 2024-11-24 17:30:00 | 45000.00 |
|      6 | Anjali  | Testing    | 2023-04-17   | 1999-12-20 | 2024-11-25 09:12:10
| 2024-11-25 18:05:00 | 47000.00 |
|      7 | Vijay   | IT         | 2021-09-14   | 1997-04-08 | 2024-11-26 09:08:15
| 2024-11-26 18:20:40 | 59000.00 |
|      8 | Divya   | HR         | 2019-10-19   | 1995-06-11 | 2024-11-27 08:50:30
| 2024-11-27 17:42:15 | 50000.00 |
|      9 | Ramesh  | Finance    | 2017-07-01   | 1991-08-14 | 2024-11-28 09:03:00
| 2024-11-28 18:12:00 | 70000.00 |
|     10 | Pooja   | Testing    | 2022-05-09   | 2001-03-19 | 2024-11-29 09:18:50
| 2024-11-29 18:25:10 | 53000.00 |

```

```
+-----+-----+-----+-----+-----+-----+
+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select emp_name, date(login_time) as login_date_only from
employee_attendance;
```

```
+-----+-----+
| emp_name | login_date_only |
+-----+-----+
| Rahul    | 2024-11-20      |
| Sneha    | 2024-11-21      |
| Arjun    | 2024-11-22      |
| Priya    | 2024-11-23      |
| Kiran    | 2024-11-24      |
| Anjali   | 2024-11-25      |
| Vijay    | 2024-11-26      |
| Divya    | 2024-11-27      |
| Ramesh   | 2024-11-28      |
| Pooja    | 2024-11-29      |
+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select emp_name, time(login_time) as login_time_only from
employee_attendance;
```

```
+-----+-----+
| emp_name | login_time_only |
+-----+-----+
| Rahul    | 09:05:10        |
| Sneha    | 08:55:20        |
| Arjun    | 09:15:45        |
| Priya    | 09:00:00        |
| Kiran    | 08:40:30        |
| Anjali   | 09:12:10        |
| Vijay    | 09:08:15        |
| Divya    | 08:50:30        |
| Ramesh   | 09:03:00        |
| Pooja    | 09:18:50        |
+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> select emp_name, year(joining_date) as joining_year from
employee_attendance;
```

```
+-----+-----+
| emp_name | joining_year |
+-----+-----+
| Rahul    | 2021         |
| Sneha    | 2020         |
| Arjun    | 2019         |
| Priya    | 2022         |
| Kiran    | 2018         |
+-----+-----+
```

Anjali	2023
Vijay	2021
Divya	2019
Ramesh	2017
Pooja	2022

+-----+-----+

10 rows in set (0.00 sec)

mysql> select emp_name, month(birth_date) as birth_month from employee_attendance;

emp_name	birth_month
----------	-------------

+-----+-----+

Rahul	5
Sneha	11
Arjun	7
Priya	1
Kiran	9
Anjali	12
Vijay	4
Divya	6
Ramesh	8
Pooja	3

+-----+-----+

10 rows in set (0.00 sec)

mysql> select emp_name, day(birth_date) as birth_day from employee_attendance;

emp_name	birth_day
----------	-----------

+-----+-----+

Rahul	10
Sneha	18
Arjun	30
Priya	25
Kiran	15
Anjali	20
Vijay	8
Divya	11
Ramesh	14
Pooja	19

+-----+-----+

10 rows in set (0.00 sec)

mysql> select emp_name, week(joining_date) as joining_week from employee_attendance;

emp_name	joining_week
----------	--------------

+-----+-----+

Rahul	24
Sneha	11
Arjun	31

Priya	1
Kiran	49
Anjali	16
Vijay	37
Divya	41
Ramesh	26
Pooja	19

+-----+

10 rows in set (0.00 sec)

```
mysql> select emp_name, dayofweek(joining_date) as joining_week_day from
employee_attendance;
```

emp_name	joining_week_day
Rahul	3
Sneha	7
Arjun	7
Priya	4
Kiran	3
Anjali	2
Vijay	3
Divya	7
Ramesh	7
Pooja	2

+-----+

10 rows in set (0.00 sec)

```
mysql> select emp_name, date_add(joining_date, interval 30 day) as
month_after_foining from employee_attendance;
```

emp_name	month_after_foining
Rahul	2021-07-15
Sneha	2020-04-20
Arjun	2019-09-09
Priya	2022-02-04
Kiran	2019-01-10
Anjali	2023-05-17
Vijay	2021-10-14
Divya	2019-11-18
Ramesh	2017-07-31
Pooja	2022-06-08

+-----+

10 rows in set (0.00 sec)

```
mysql> select emp_name, date_sub(joining_date, interval 15 day) as
15days_before_joining from employee_attendance;
```

emp_name	15days_before_joining
----------	-----------------------

Rahul	2021-05-31
Sneha	2020-03-06
Arjun	2019-07-26
Priya	2021-12-21
Kiran	2018-11-26
Anjali	2023-04-02
Vijay	2021-08-30
Divya	2019-10-04
Ramesh	2017-06-16
Pooja	2022-04-24

10 rows in set (0.00 sec)

```
mysql> select emp_name, datediff(joining_date, now()) as experience from
employee_attendance;
```

emp_name	experience
Rahul	-1919
Sneha	-2370
Arjun	-2594
Priya	-1715
Kiran	-2836
Anjali	-1248
Vijay	-1828
Divya	-2524
Ramesh	-3364
Pooja	-1591

10 rows in set (0.00 sec)

```
mysql> select emp_name, datediff(now(), joining_date) as experience from
employee_attendance;
```

emp_name	experience
Rahul	1919
Sneha	2370
Arjun	2594
Priya	1715
Kiran	2836
Anjali	1248
Vijay	1828
Divya	2524
Ramesh	3364
Pooja	1591

10 rows in set (0.00 sec)

```
mysql> select emp_name, timediff(logout_time, login_time) as work_hours from
employee_attendance;
```

emp_name	work_hours
Rahul	09:10:20
Sneha	08:50:30
Arjun	09:19:35
Priya	08:55:40
Kiran	08:49:30
Anjali	08:52:50
Vijay	09:12:25
Divya	08:51:45
Ramesh	09:09:00
Pooja	09:06:20

10 rows in set (0.00 sec)

```
mysql> select emp_name, date_format(joining_date, '%D, %M %d, %Y') as work_hours
from employee_attendance;
```

emp_name	work_hours
Rahul	15th, June 15, 2021
Sneha	21st, March 21, 2020
Arjun	10th, August 10, 2019
Priya	5th, January 05, 2022
Kiran	11th, December 11, 2018
Anjali	17th, April 17, 2023
Vijay	14th, September 14, 2021
Divya	19th, October 19, 2019
Ramesh	1st, July 01, 2017
Pooja	9th, May 09, 2022

10 rows in set (0.00 sec)

```
mysql> select emp_name, date_format(joining_date, '%W, %M %d, %Y') as work_hours
from employee_attendance;
```

emp_name	work_hours
Rahul	Tuesday, June 15, 2021
Sneha	Saturday, March 21, 2020
Arjun	Saturday, August 10, 2019
Priya	Wednesday, January 05, 2022
Kiran	Tuesday, December 11, 2018
Anjali	Monday, April 17, 2023
Vijay	Tuesday, September 14, 2021
Divya	Saturday, October 19, 2019
Ramesh	Saturday, July 01, 2017

Pooja	Monday, May 09, 2022
-------	----------------------

10 rows in set (0.00 sec)

```
mysql> select emp_name from employee_attendance where year(joining_date) = 2026;
Empty set (0.00 sec)
```

```
mysql> select emp_name from employee_attendance where month(joining_date) = 11;
Empty set (0.00 sec)
```

```
mysql> select emp_name from employee_attendance where month(birth_date) = 11;
+-----+
| emp_name |
+-----+
| Sneha    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select emp_name from employee_attendance where month(joining_date) = 06;
+-----+
| emp_name |
+-----+
| Rahul    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select emp_name from employee_attendance where day(birth_date) = 10;
+-----+
| emp_name |
+-----+
| Rahul    |
+-----+
1 row in set (0.00 sec)
```

```
mysql> select emp_name from employee_attendance where quarter(joining_date) = 2;
+-----+
| emp_name |
+-----+
| Rahul    |
| Anjali   |
| Pooja    |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> select emp_name, timestampdiff(year,date(now()),birth_date) as age from
employee_attendance;
+-----+-----+
| emp_name | age |
+-----+-----+
| Rahul    | -28 |
+-----+-----+
```

Sneha	-29
Arjun	-32
Priya	-26
Kiran	-34
Anjali	-26
Vijay	-29
Divya	-31
Ramesh	-35
Pooja	-25

10 rows in set (0.00 sec)

```
mysql> select emp_name, timestampdiff(year,birth_date,curdate()) as age from
employee_attendance;
```

emp_name	age
Rahul	28
Sneha	29
Arjun	32
Priya	26
Kiran	34
Anjali	26
Vijay	29
Divya	31
Ramesh	35
Pooja	25

10 rows in set (0.00 sec)

```
mysql> select emp_name from employee_attendance where
timestampdiff(year,joining_date,curdate()) > 03;
```

emp_name
Rahul
Sneha
Arjun
Priya
Kiran
Vijay
Divya
Ramesh
Pooja

9 rows in set (0.00 sec)

```
mysql> select emp_name,timestampdiff(hour,login_time,logout_time) as working_hours
from employee_attendance;
```

+-----+

emp_name	working_hours
Rahul	9
Sneha	8
Arjun	9
Priya	8
Kiran	8
Anjali	8
Vijay	9
Divya	8
Ramesh	9
Pooja	9

10 rows in set (0.00 sec)

```
mysql> select emp_name,timestampdiff(minute,login_time,logout_time) as
working_minutes from employee_attendance;
```

emp_name	working_minutes
Rahul	550
Sneha	530
Arjun	559
Priya	535
Kiran	529
Anjali	532
Vijay	552
Divya	531
Ramesh	549
Pooja	546

10 rows in set (0.00 sec)

```
mysql> select emp_name from employee_attendance where time(login_time) < '09:00:00'
-> ;
```

emp_name
Sneha
Kiran
Divya

3 rows in set (0.00 sec)

```
mysql> select emp_name from employee_attendance where time(login_time) >
'09:10:00';
```

emp_name
Arjun

Anjali
Pooja

+-----+

3 rows in set (0.00 sec)

```
mysql> select emp_name from employee_attendance where dayname(joining_date) =
'Monday';
```

+-----+

emp_name

+-----+

Anjali

Pooja

+-----+

2 rows in set (0.00 sec)

```
mysql> select emp_name, birth_date from employee_attendance where
monthname(birth_date) like 'j%';
```

+-----+-----+

emp_name	birth_date
----------	------------

+-----+-----+

Arjun	1994-07-30
-------	------------

Priya	2000-01-25
-------	------------

Divya	1995-06-11
-------	------------

+-----+-----+

3 rows in set (0.03 sec)

```
mysql> select emp_name, joining_date from employee_attendance where
week(joining_date) = 25;
```

Empty set (0.00 sec)

```
mysql> select emp_name, last_day(birth_date) from employee_attendance;
```

+-----+-----+

emp_name	last_day(birth_date)
----------	----------------------

+-----+-----+

Rahul	1998-05-31
-------	------------

Sneha	1996-11-30
-------	------------

Arjun	1994-07-31
-------	------------

Priya	2000-01-31
-------	------------

Kiran	1992-09-30
-------	------------

Anjali	1999-12-31
--------	------------

Vijay	1997-04-30
-------	------------

Divya	1995-06-30
-------	------------

Ramesh	1991-08-31
--------	------------

Pooja	2001-03-31
-------	------------

+-----+-----+

10 rows in set (0.00 sec)

```
mysql> notee
```